

Chapter 3

Continuous Probability Distributions

In Chapter 2 we built probability histograms: bars of width 1 whose heights summed to 1, so that probability equaled area. We watched the binomial histogram grow more bell-shaped as n increased, and we promised that the bars were becoming a curve.

A **continuous probability distribution** is what you get when you let the number of bars grow without bound while keeping total area equal to 1. The bars become infinitely thin and infinitely numerous. Their tops trace a smooth **density curve**. Individual bars disappear, but any region under the curve still has a perfectly well-defined area, and that area is still probability.

Every statistical inference technique from Chapter 5 onward rests on computing areas under curves. We will learn two such curves in this chapter: the simplest one (the uniform distribution) and the most important one in all of statistics (the standard normal distribution).

3.1 From Bars to Curves: The Density Function

Definition: Probability Density Function. A probability density function (pdf) is a curve $f(x)$ such that:

1. $f(x) \geq 0$ for all x (the curve lies on or above the horizontal axis).
2. The total area under the curve equals 1.

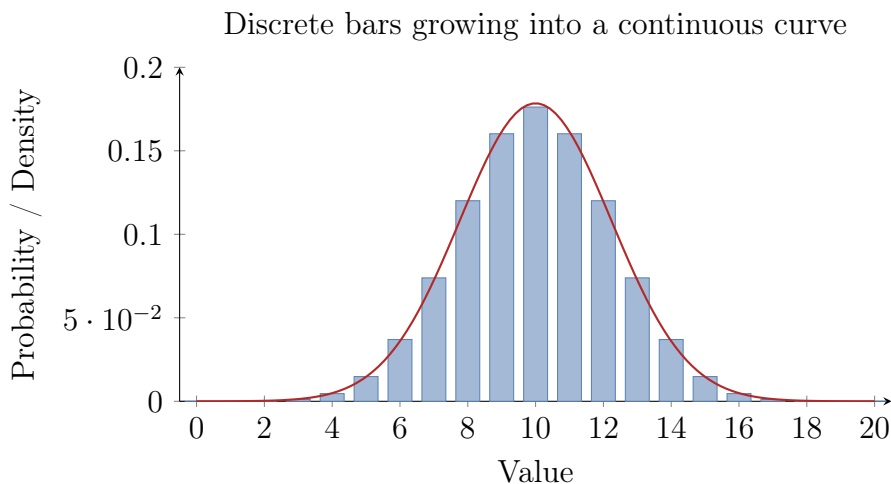
For any interval $[a, b]$, the probability that X falls in that interval equals the area under the curve between a and b :

$$P(a \leq X \leq b) = \text{area under } f(x) \text{ between } a \text{ and } b.$$

Visual Insight. This is the same rule as Chapter 2, just extended. Before: probability was bar area. Now: probability is area under a curve. T

Here is the transition visualized: a discrete binomial histogram ($n = 20, p = 0.5$)

with a smooth normal curve drawn over it. The bars are already approximating the curve. As $n \rightarrow \infty$, the bars vanish and only the curve remains.



An important consequence. For a continuous distribution, the probability of any single exact value is zero:

$$P(X = c) = 0 \quad \text{for any specific value } c.$$

This is because a single point has no width, and therefore no area. As a result,

$$P(a \leq X \leq b) = P(a < X < b)$$

for continuous distributions; including or excluding the endpoints makes no difference.

Exercises 3.1

- 3.1.1. Explain in your own words why $P(X = 2.5) = 0$ for a continuous random variable, but $P(X = 2.5)$ is perfectly meaningful for a discrete random variable.
- 3.1.2. A density curve has a triangular shape: it rises linearly from $x = 0$ to $x = 2$, reaching a peak, then drops back to 0 at $x = 4$. Without computing, where must the peak be? What must its height be?
- 3.1.3. True or False: For a continuous distribution, $P(X > 3)$ and $P(X \geq 3)$ are different numbers. Explain.

3.2 The Uniform Distribution

The simplest continuous distribution is one where every value in a range is equally likely.

Definition: Uniform Distribution. A random variable X has a **uniform distribution** on the interval $[a, b]$ (written $X \sim U(a, b)$) if its density function is the flat, constant curve:

$$f(x) = \frac{1}{b-a} \quad \text{for } a \leq x \leq b, \quad f(x) = 0 \text{ otherwise.}$$

The height $\frac{1}{b-a}$ is chosen so that the total area (a rectangle of width $b-a$ and height $\frac{1}{b-a}$) equals exactly 1.

Key Formula. For $X \sim U(a, b)$:

$$\mu = \frac{a+b}{2}, \quad \sigma = \frac{b-a}{\sqrt{12}}.$$

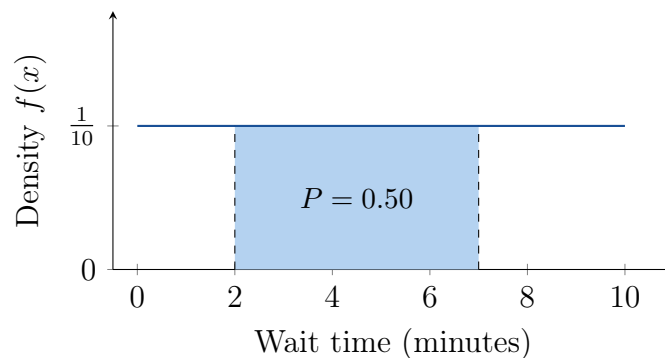
For any subinterval $[c, d] \subseteq [a, b]$, the probability is:

$$P(c \leq X \leq d) = \frac{d-c}{b-a} = \frac{\text{width of interval}}{\text{total width}}.$$

The probability formula is just the area of a rectangle: width times height, which is $(d-c) \cdot \frac{1}{b-a}$.

Example 3.2.1. (Bus Arrival Time) A city bus arrives at a stop uniformly between 0 and 10 minutes after the hour. $X \sim U(0, 10)$. The density is $f(x) = \frac{1}{10}$ on $[0, 10]$.

$U(0, 10)$: Bus arrival



What is the probability the bus arrives between 2 and 7 minutes after the hour?

$$P(2 \leq X \leq 7) = \frac{7-2}{10-0} = \frac{5}{10} = 0.50.$$

The shaded rectangle has width 5 and height $\frac{1}{10}$: area = 0.50.

What is the probability the bus arrives within the first 3 minutes?

$$P(0 \leq X \leq 3) = \frac{3}{10} = 0.30.$$

Mean: $\mu = \frac{0+10}{2} = 5$ minutes. On average, you wait 5 minutes.

Example 3.2.2. (*Uniform: Complement*) Same bus. What is the probability you wait more than 8 minutes?

$$P(X > 8) = \frac{10 - 8}{10} = \frac{2}{10} = 0.20.$$

Or equivalently, $P(X > 8) = 1 - P(X \leq 8) = 1 - 0.80 = 0.20$. The complement rule always works, and here it gives a nice check.

Exercises 3.2

3.2.1. A random number generator produces values uniformly on $[0, 1]$. Compute: (a) $P(X < 0.3)$; (b) $P(0.2 < X < 0.8)$; (c) $P(X > 0.95)$.

3.2.2. For $X \sim U(5, 20)$, compute μ , σ , and $P(8 \leq X \leq 15)$. Sketch the density curve and shade the region for $P(8 \leq X \leq 15)$.

3.2.3. A flight's departure time is uniformly distributed between 0 and 30 minutes late. What is the probability it departs more than 20 minutes late? What is the expected delay?

3.3 The Standard Normal Distribution

The normal distribution is the most important distribution in all of statistics. It describes the distribution of averages, measurement errors, test scores, heights, and countless other phenomena. More importantly for us, it is the foundation of all the hypothesis tests we will build from Chapter 5 onward.

3.3.1 The Bell Curve

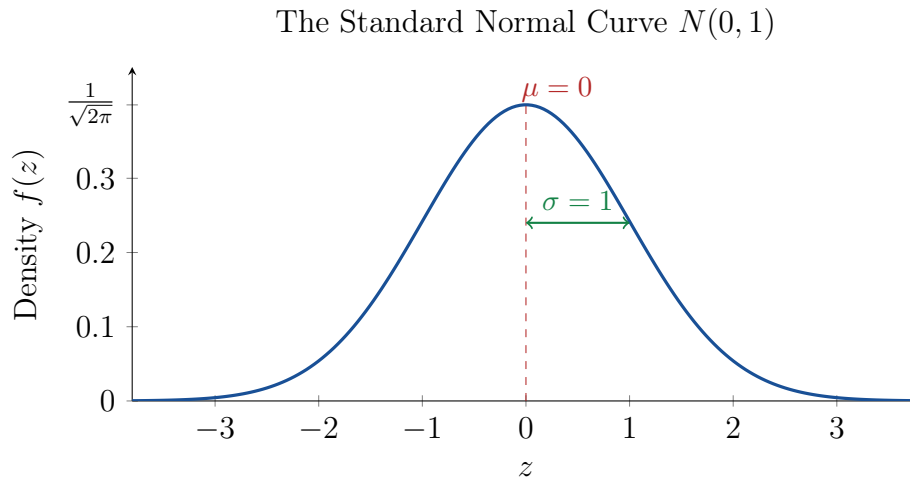
Definition: Standard Normal Distribution. The standard normal distribution, written $Z \sim N(0, 1)$, has:

- Mean $\mu = 0$.
- Standard deviation $\sigma = 1$.

- Density function:

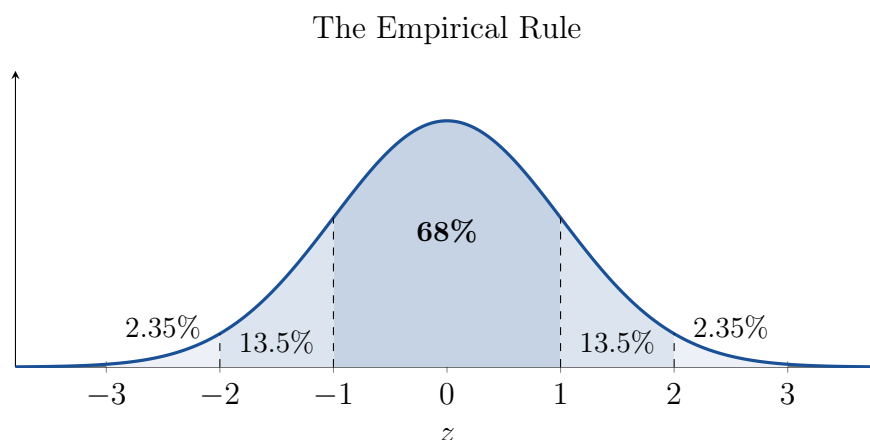
$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

Do not be intimidated by the formula for $f(z)$. You will not compute it by hand. What matters is the shape and the properties.



3.3.2 Properties of the Standard Normal

1. **Symmetric** around $z = 0$. The left and right halves are mirror images.
2. **Bell-shaped**: highest at the center, tapering toward the tails.
3. **Total area** = 1. Always.
4. The curve approaches but never touches the horizontal axis (the tails are *asymptotic*).
5. **The Empirical Rule** (also called the 68-95-99.7 rule):
 - Approximately 68% of area lies within $\pm 1\sigma$ of the mean.
 - Approximately 95% of area lies within $\pm 2\sigma$ of the mean.
 - Approximately 99.7% of area lies within $\pm 3\sigma$ of the mean.



Exercises 3.3

3.3.1. For $Z \sim N(0, 1)$, use the Empirical Rule to estimate (no table needed):
 (a) $P(-1 \leq Z \leq 1)$; (b) $P(Z > 2)$; (c) $P(-3 \leq Z \leq 0)$.

3.3.2. Why does the standard normal curve never touch the horizontal axis, yet the total area under it is still exactly 1?

3.3.3. Sketch the standard normal curve. Shade the region that represents 95% of the area. Between what two z -values does it lie?

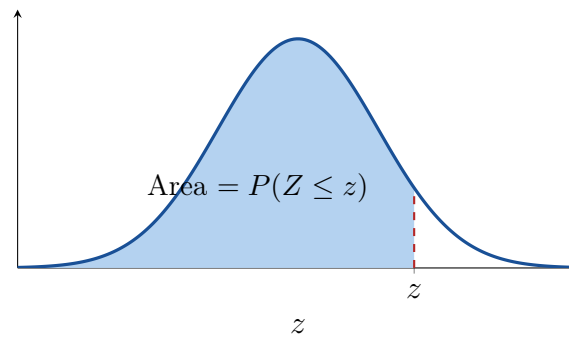
3.4 The z -Table: Reading Probability as Area

We need to compute exact probabilities — areas under the normal curve — for values other than those covered by the Empirical Rule. A **z -table** (also called the standard normal table) provides these areas.

3.4.1 What the Table Gives You

The z -table gives the area **to the left** of a specified z -value. That is, it gives $P(Z \leq z)$ for any z you look up.

The table gives this area: $P(Z \leq z)$

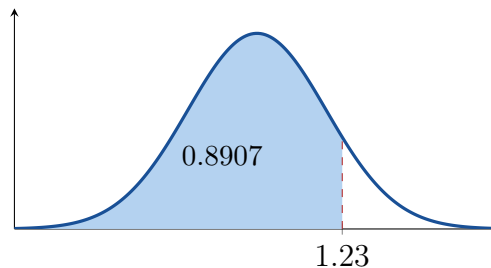


3.4.2 Using the Table

A z -value is written to two decimal places. The table is organized by the ones-and-tenths digit (row) and the hundredths digit (column). For example, to find $P(Z \leq 1.45)$: find the row for 1.4, then the column for 0.05. The entry is **0.9265**.

Example 3.4.1. (*Left-Tail Probability*) Find $P(Z \leq 1.23)$.

Look up row 1.2, column 0.03: the table gives **0.8907**.



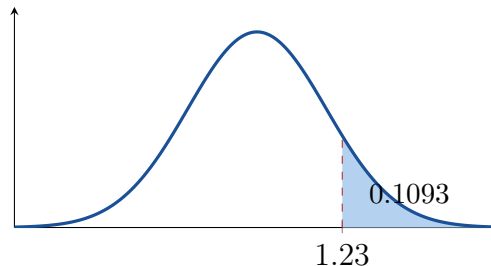
3.4.3 Right-Tail Probabilities

The table gives left-tail areas. For a right-tail probability, use the complement rule:

$$P(Z > z) = 1 - P(Z \leq z).$$

Example 3.4.2. (*Right-Tail Probability*) Find $P(Z > 1.23)$.

$$P(Z > 1.23) = 1 - P(Z \leq 1.23) = 1 - 0.8907 = 0.1093.$$



3.4.4 Between Two Values

For a probability between two z -values, subtract two table entries:

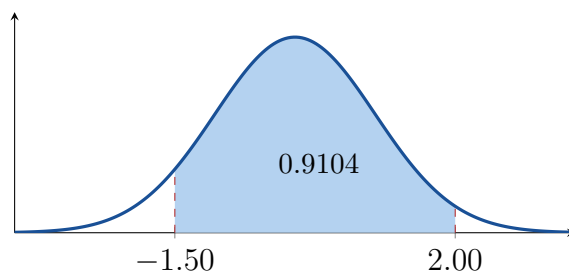
$$P(a \leq Z \leq b) = P(Z \leq b) - P(Z \leq a).$$

Think of it as: big left area minus small left area equals the middle strip.

Example 3.4.3. (*Between Two z -Values*) Find $P(-1.50 \leq Z \leq 2.00)$.

$$P(Z \leq 2.00) = 0.9772, \quad P(Z \leq -1.50) = 0.0668.$$

$$P(-1.50 \leq Z \leq 2.00) = 0.9772 - 0.0668 = 0.9104.$$



3.4.5 Negative z -Values

The table includes negative z -values. Because the curve is symmetric, $P(Z \leq -z) = 1 - P(Z \leq z)$. Always confirm this makes intuitive sense by sketching.

Common Pitfall. Always draw the picture first. Before looking anything up, sketch the normal curve, mark the z -value(s), shade the region you want, and estimate whether the answer should be bigger or smaller than 0.5. The picture tells you whether to use the table value directly, subtract from 1, or subtract two table values. Students who skip the picture make errors; students who draw the picture catch them.

Example 3.4.4. (*Negative z -Value*) Find $P(Z \leq -0.85)$.

Look up $z = -0.8$, column 0.05: the table gives **0.1977**.

Sanity check: -0.85 is left of center. Less than half the area is to the left. $0.1977 < 0.50$. ✓

Example 3.4.5. (*Working Backwards: Finding z from a Probability*) What z -value has 90% of the distribution below it?

We want $P(Z \leq z) = 0.90$. Search the body of the z -table for 0.9000 (or closest entry). We find 0.8997 at $z = 1.28$ and 0.9015 at $z = 1.29$. We typically use $z = 1.28$.

Exercises 3.4

- 3.4.1. Using a z -table, find: (a) $P(Z \leq 0.54)$; (b) $P(Z \leq -1.32)$; (c) $P(Z > 2.10)$; (d) $P(-0.75 \leq Z \leq 1.50)$. For each, draw the normal curve and shade the appropriate region.
- 3.4.2. Find the z -value such that: (a) 10% of the distribution is below it; (b) 5% of the distribution is above it.
- 3.4.3. By symmetry, $P(Z \leq -1.96) = P(Z \geq 1.96)$. Look up both values and confirm they are equal. What is $P(-1.96 \leq Z \leq 1.96)$? Why is the value 1.96 important? (Preview: it will appear again in Chapter 5.)

3.5 Area as Probability: The Unifying Thread

We have now seen three distributions, and in each case probability has been area:

Distribution	Shape	How to find $P(a \leq X \leq b)$
Discrete (e.g., Binomial)	Bars	Sum bar heights (or areas)
Uniform $U(a, b)$	Rectangle	$\frac{b-a}{b-a}$ (rectangle area)
Standard Normal $N(0, 1)$	Bell curve	z -table

In Chapter 4, we will extend the standard normal to handle any normal distribution with mean μ and standard deviation σ , not just the special case $\mu = 0$, $\sigma = 1$. The bridge from any normal distribution to the standard normal table is a single formula called the z -score. It converts any value to a standard unit that the table can read.

Chapter 3 Review

- A coffee machine dispenses a uniformly distributed amount of coffee between 7 and 9 ounces per cup. (a) Write the density function. What is the height of the uniform curve? (b) Sketch the density curve and find $P(7.5 \leq X \leq 8.5)$ by computing a rectangle area. (c) Find μ and σ . (d) What is $P(X > 8.8)$?
- Using a z -table, compute and sketch: (a) $P(Z \leq 1.65)$; (b) $P(Z > -0.42)$; (c) $P(1.00 \leq Z \leq 2.33)$; (d) $P(|Z| \leq 1.96)$ (i.e., $P(-1.96 \leq Z \leq 1.96)$).
- Find the z^* value such that: (a) $P(Z \leq z^*) = 0.025$; (b) $P(Z > z^*) = 0.025$. These two values will appear throughout the rest of the course.

4. A student takes a very difficult exam. Scores follow a standard normal distribution ($\mu = 0$, $\sigma = 1$ after standardizing). (a) What fraction of students scored above $z = 1.5$? (b) Between $z = -1$ and $z = 1$? (c) The professor passes the top 15% of students. What is the minimum z -score needed to pass?
5. *Big picture*: In three sentences, explain the connection between the binomial probability histogram from Chapter 2 and the standard normal density curve from this chapter. Why is the normal distribution the natural “limit” of the binomial?

Standard Normal Table

This table gives left-tail areas for the standard normal distribution. Each entry is $P(Z \leq z)$, the area to the left of the given z -value. Use the row for the ones-and-tenths place and the column for the hundredths place. For example, row 1.2 and column 0.03 give $P(Z \leq 1.23) = 0.8907$. For negative values, use symmetry:

$$P(Z \leq -z) = 1 - P(Z \leq z).$$

So, for example, $P(Z \leq -1.23) = 1 - 0.8907 = 0.1093$.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998